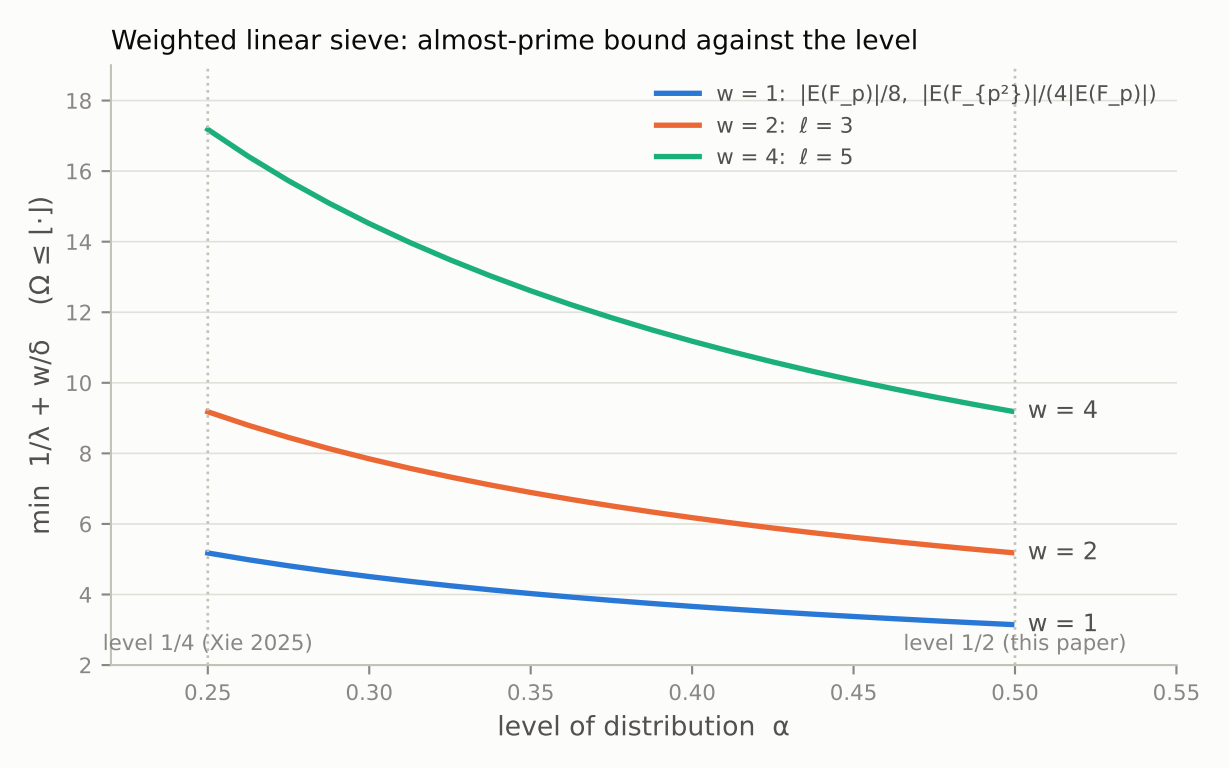
Weighted linear sieve: almost-prime bound against the level
w = 1: |E(F_p)|/8, |E(F_{p²})|/(4|E(F_p)|)
w = 2: ℓ = 3
w = 4: ℓ = 5
min 1/λ + w/δ (Ω ≤ ⌊·⌋)
level of distribution α
w = 4
w = 2
w = 1
level 1/4 (Xie 2025)
level 1/2 (this paper)
18
16
14
12
10
8
6
4
2
0.25
0.30
0.35
0.40
0.45
0.50
0.55